

BOTEV VRAH, Bulgaria

WMO: 156270

Lat: 42.7170N

Lon: 24.9170E

Elev: 2389

StdP: 75.73

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.5	-17.8	-24.2	0.6	-13.8	-21.9	0.7	-13.2	34.4	-9.5	34.2	-9.4	12.3	320	1.073

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.1	16.1	10.3	14.6	9.8	13.4	9.4	11.9	14.0	11.1	13.2	10.2	12.3	5.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	11.1	12.8	10.2	10.4	11.8	9.4	9.8	11.0	41.4	14.0	38.9	13.3	36.6	12.4	16.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 34.0	28.2	24.0	DB	-21.5	18.7	2.7	1.4	-23.5	19.7	-25.1	20.6	-26.7	21.3	-28.7	22.4	(4)
(5)			WB	-21.9	13.8	2.7	1.0	-23.9	14.5	-25.5	15.1	-27.0	15.6	-29.0	16.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-7.9	-8.1	-6.0	-2.0	3.1	6.5	8.6	9.0	5.2	2.0	-2.0	-6.1
	DBStd	7.38	4.83	4.93	4.67	3.66	3.49	3.63	3.13	2.99	3.54	4.39	5.30	4.94
	HDD10.0	3622	555	506	495	359	215	116	64	55	148	250	359	499
	HDD18.3	6603	813	739	753	609	473	357	303	289	394	507	609	758
	CDD10.0	60	0	0	0	0	1	9	21	24	4	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.9	10.5	10.7	11.0	9.6	8.1	7.8	7.2	6.4	7.3	8.4	9.9	10.3
Precipitation	PrecAvg	804	55	50	56	67	101	103	101	65	54	47	39	64
	PrecMax	1237	134	90	133	142	200	213	268	189	167	124	100	127
	PrecMin	591	14	17	27	24	14	19	7	1	6	4	5	18
	PrecStd	162	29	18	25	31	52	49	72	48	37	28	20	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	3.3	6.4	8.2	12.8	16.6	18.7	18.0	14.6	13.1	10.0	5.3
		MCWB	-0.4	-2.0	0.2	4.3	7.1	10.3	11.1	11.7	8.6	6.5	2.8	0.5
	2%	DB	1.0	0.7	3.5	6.3	10.5	14.1	16.2	16.1	12.8	10.6	8.2	3.1
		MCWB	-2.7	-2.3	-0.4	3.3	6.3	9.8	10.1	10.4	7.8	5.3	2.0	-0.6
	5%	DB	-0.6	-0.8	1.6	4.1	8.9	12.8	14.4	14.8	11.4	9.1	6.4	1.7
		MCWB	-2.7	-2.8	-1.3	1.9	5.9	9.3	10.1	9.9	7.2	4.5	0.9	-1.0
	10%	DB	-2.0	-2.2	0.1	2.4	7.7	11.4	12.9	13.4	10.0	7.5	4.6	0.2
		MCWB	-3.5	-3.5	-1.6	0.8	5.4	8.8	9.6	9.6	7.0	3.7	0.4	-2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	-0.3	2.4	5.6	8.8	12.4	13.0	13.5	10.4	8.2	5.5	3.0
		MCDB	1.3	0.9	4.6	7.3	10.7	13.9	15.2	16.0	12.8	10.1	7.8	3.7
	2%	WB	-1.2	-1.6	0.4	3.7	7.4	11.2	11.8	12.1	9.1	6.7	3.8	1.0
		MCDB	0.1	-0.2	2.0	5.7	8.9	12.8	14.1	14.1	11.5	9.0	6.1	2.0
	5%	WB	-2.3	-2.5	-0.6	2.2	6.5	10.3	10.9	11.1	8.2	5.9	2.4	-0.5
		MCDB	-1.2	-1.3	0.7	3.4	8.2	11.9	13.1	13.4	10.2	7.8	3.8	0.8
	10%	WB	-3.3	-3.4	-1.5	1.1	5.7	9.2	10.1	10.1	7.5	4.9	1.3	-1.7
		MCDB	-2.4	-2.3	-0.1	2.2	7.3	10.9	12.1	12.4	9.2	6.4	3.6	-0.8
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.3	4.3	3.7	3.6	3.8	4.1	4.1	3.8	4.0	4.1	4.2
		MCDBR	5.0	5.1	4.7	4.4	4.4	5.3	5.9	5.2	5.4	4.7	4.3	4.7
		MCWBR	3.8	4.2	3.7	4.0	4.3	4.9	4.6	4.7	4.4	3.9	3.3	3.7
	5% WB	MCDBR	4.3	4.3	4.3	4.1	4.1	4.3	4.9	4.8	4.3	3.8	3.6	4.4
		MCWBR	3.8	4.1	3.8	4.0	4.1	4.9	4.8	4.8	4.1	3.9	3.7	4.0
Clear-Sky Solar Irradiance	taub	0.247	0.268	0.299	0.337	0.327	0.329	0.331	0.337	0.318	0.291	0.269	0.251	
	taud	2.465	2.445	2.422	2.314	2.386	2.390	2.379	2.360	2.413	2.494	2.531	2.491	
	Ebn at Noon	918	947	949	929	945	940	934	918	915	905	883	882	
	Edh at Noon	71	87	101	123	117	117	117	116	101	82	67	64	
All-Sky Solar Radiation	RadAvg	1.58	2.38	3.43	4.58	5.60	6.16	6.37	5.76	4.26	2.82	1.77	1.35	
	RadStd	0.23	0.31	0.31	0.50	0.45	0.47	0.47	0.40	0.33	0.32	0.19	0.18	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	+0.69	+0.82	N/A	N/A	N/A
(47)	Regional (37 neighbors)	+0.45	N/A	N/A	N/A	+0.53	+0.50	N/A	-142	N/A

Nomenclature: See separate page